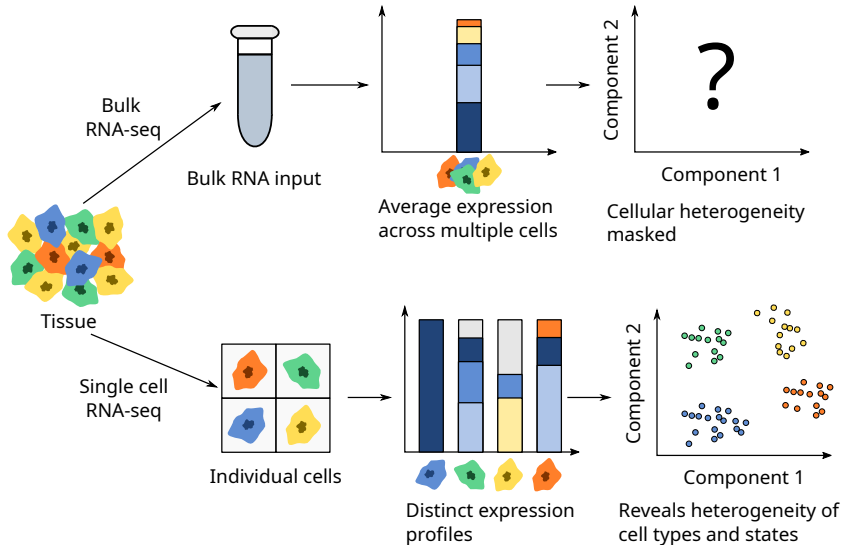




Error Correction Algorithms for Efficient Gene Expression Quantification in Single Cell Transcriptomics

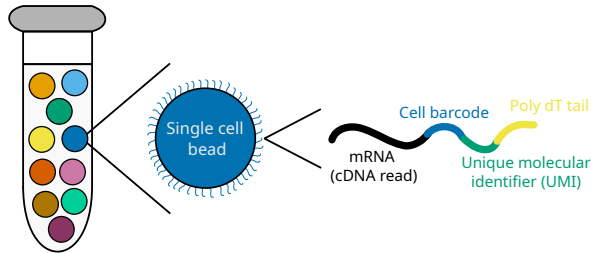
Jens Zentgraf, Johanna E. Schmitz, Andreas Keller, Sven Rahmann

Motivation: RNAseq



Motivation

- Massive increase in data
- Possible to sequence thousands of cells in parallel
- 1st step:
Create cell x gene count matrix
- Faster and scalable methods needed



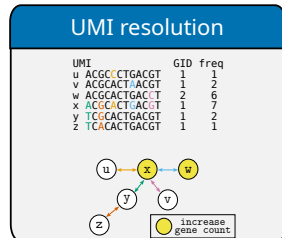
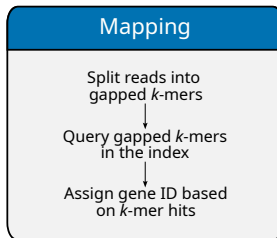
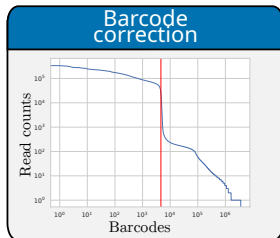
@Header
GTAACGAT
+
FFFFFFFF
R1

Barcode + UMI
ACTGACTGACAAGTCTGTGTGACAAGAT
16 + 12 base pairs

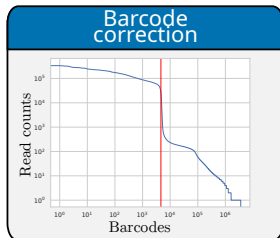
@Header
ACGCATG
+
FFFFFFFF
R2

mRNA sequence
ACTGAAACGTGATGACATG.....TGAT
100 base pairs

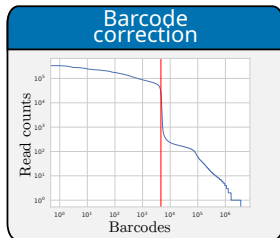
Problem



Problem



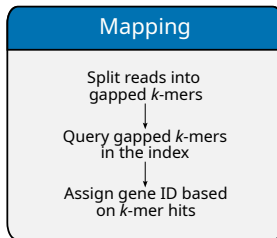
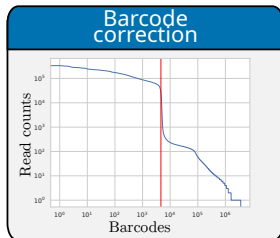
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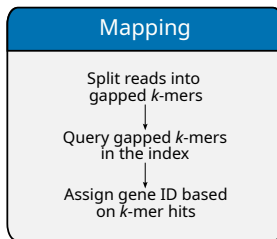
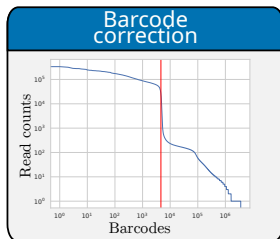
Barcode correction

- Each cell has a unique barcode from an inclusion list
- Some droplets contain no cell or a corrupted cell
- Sequencing errors lead to wrong barcodes
- Goal:
 - Discard empty droplets
 - Rescue barcodes with sequencing errors

Problem



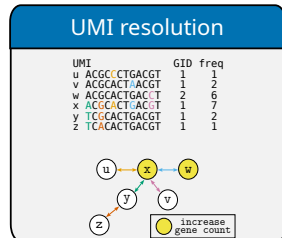
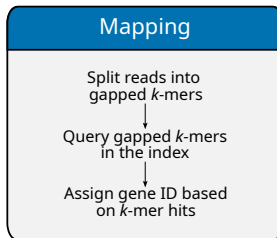
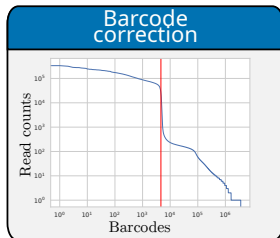
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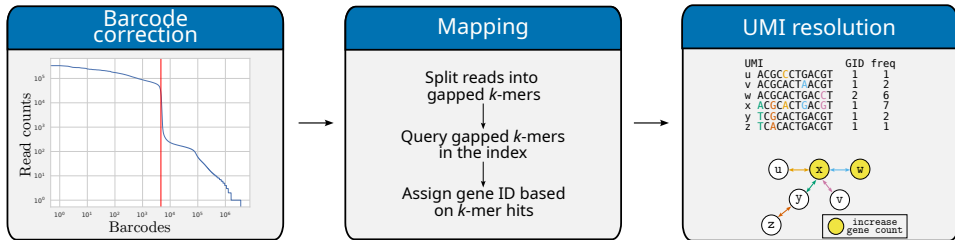
Mapping

- Map reads to the correct gene
- Similar or overlapping genes
- Avoid alignment (slow)
- Alignment-free approach

Problem



Problem



UMI resolution

- Each UMI corresponds in principle to a unique molecule
- Same UMIs should be counted only once
- Sequencing errors can inflate the number of distinct UMIs
- UMI collision can cause the same UMI to map to different genes
- Differentiate between true UMI collisions and mapping errors

Background

Definitions

- k -mers
 - Substrings of length k

ACTGAAGCTAGCT
ACTGA
CTGAA
TGAAG
GAAGC
AAGCT
AGCTA
GCTAG
CTAGC
TAGCT

Zentgraf, J., Rahmann, S.: Swiftly identifying strongly unique k -mers. Algorithms for Molecular Biology 20(1), 13 (2025). <https://doi.org/10.1186/s13015-025-00286-6>

Background

Definitions

- k -mers
 - Substrings of length k
- Gapped k -mers (spaced seeds)
 - k significant positions (#)
 - Window size w
 - $w - k$ insignificant positions (_)
 - ##_#_##

```
ACTGAAGCTAGCT
AC  G  AG
CT  A  GC
TG  A  CT
GA  G  TA
AA  C  AG
AG  T  GC
GC  A  CT
```

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- A k -mer $x \in K$ is weak in K iff a k -mer $y \in K$ with a Hamming distance of $H(x, y) = 1$ exists in K . The other k -mers are strong in K .

```
TTACTGATTAGTT
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GTAGTACGTAGGA
ACTGAGGCTAGCT
CCTACACCTTGCA
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```
TTACTGATTAGTT
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ACTGAGGCTAGCT
CCTACACCTTGCA
```

- Strongly unique k -mers are robust identifiers.

Zentgraf, J., Rahmann, S.: Swiftly identifying strongly unique k -mers. Algorithms for Molecular Biology 20(1), 13 (2025). <https://doi.org/10.1186/s13015-025-00286-6>

State of the art

CellRanger

- Commercial solution from 10X Genomics
 - Splicing-aware alignment to the reference genome
 - Uses STAR aligner
 - Slow
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- Alignment-free
- Pseudoalignment using a color-compacted de Bruijn graph

[1] Melsted, P., Boeshaghi, A.S., Liu, L., Gao, F., Lu, L., Min, K.H.J., da Veiga Beltrame, E., Hjørleifsson, K.E., Gehring, J., Pachter, L.: Modular, efficient and constant-memory single-cell RNA-seq preprocessing. *Nature Biotechnology* 39(7), 813–818 (2021). <https://doi.org/10.1038/s41587-021-00870-2>

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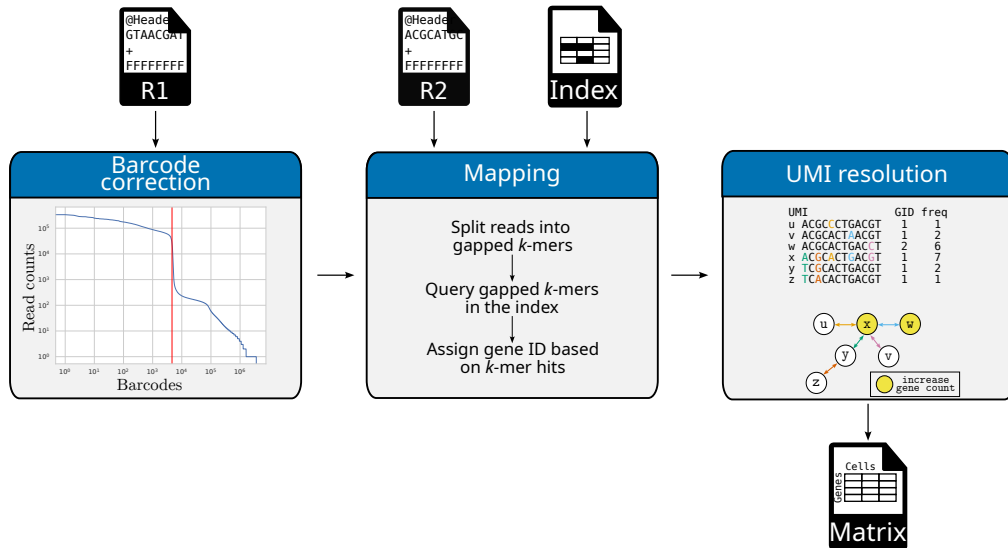
Alevin-Fry²

- Alignment-free
- Pseudoalignment using a color-compacted de Bruijn graph
- CR-like or custom barcode correction and UMI resolution

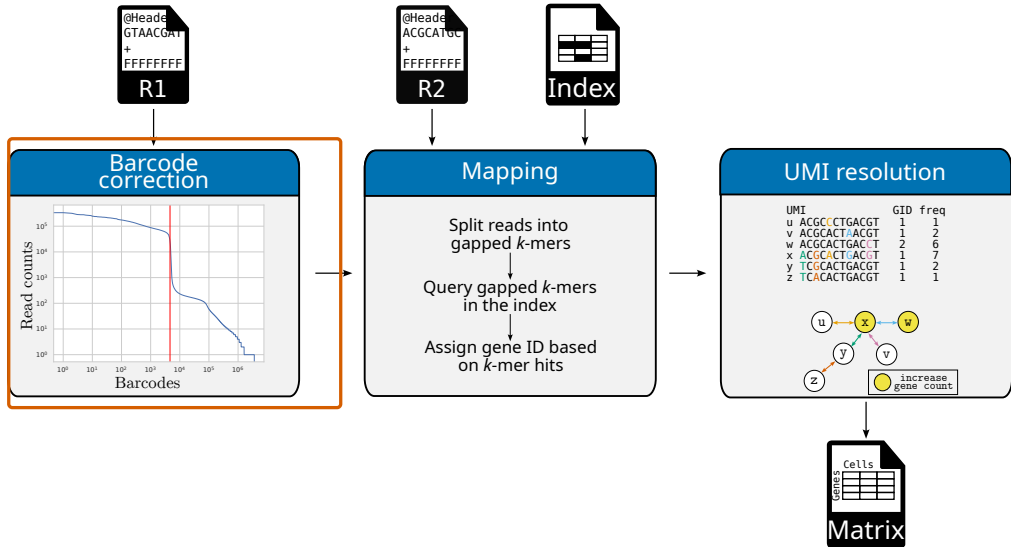
[1] Melsted, P., Boeshaghi, A.S., Liu, L., Gao, F., Lu, L., Min, K.H.J., da Veiga Beltrame, E., Hjørleifsson, K.E., Gehring, J., Pachter, L.: Modular, efficient and constant-memory single-cell RNA-seq preprocessing. *Nature Biotechnology* 39(7), 813–818 (2021). <https://doi.org/10.1038/s41587-021-00870-2>

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Overview of arcane



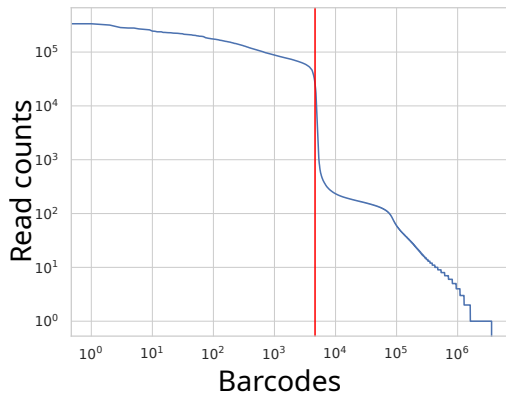
Overview of arcane



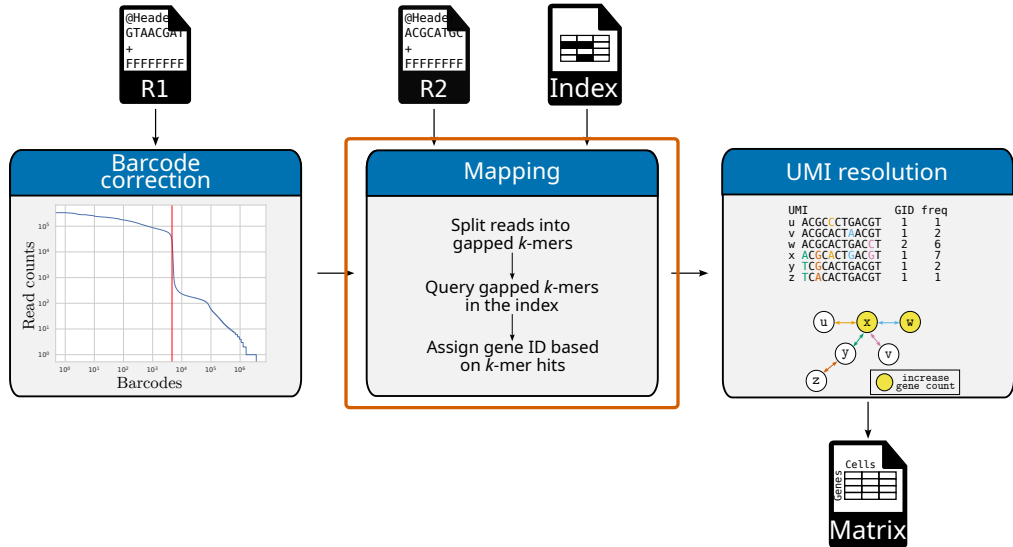
Barcode correction

Solution

- Correct invalid barcodes that have a Hamming distance 1 neighbor on the inclusion list
- Discard all barcodes with a count below the *Knee* (red line)



Overview of arcane

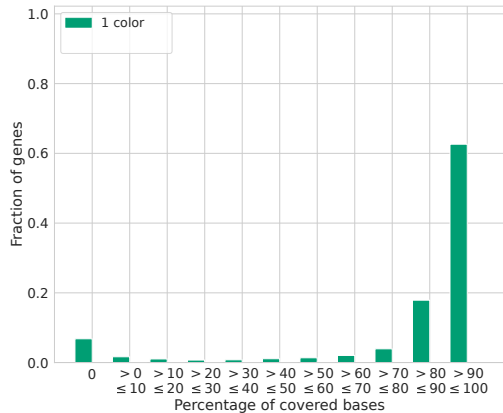


Arcane index

- Other tools use color-compacted de Bruijn graphs
- Do we need all colors?
 - More colors, more contradictory data

Arcane index

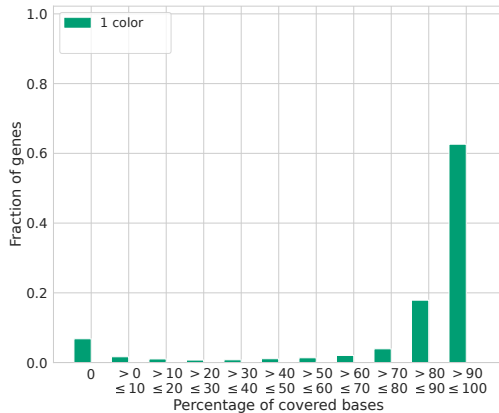
- Other tools use color-compacted de Bruijn graphs
- Do we need all colors?
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Arcane index

Observation

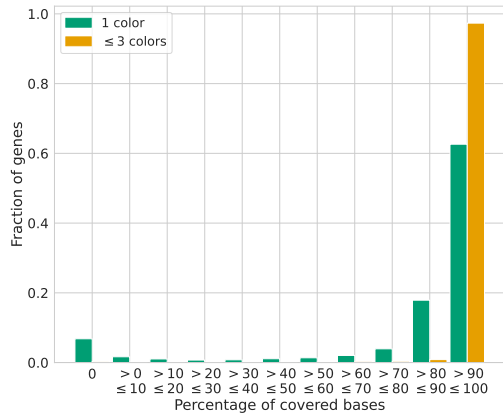
- Already high coverage using one color



Arcane index

Observation

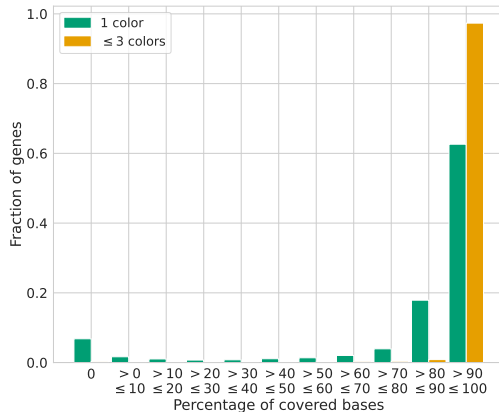
- Already high coverage using one color



Arcane index

Observation

- Already high coverage using one color
- Most genes are covered using up to 3 colors



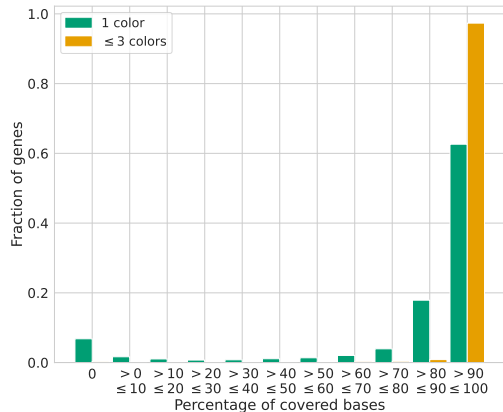
Arcane index

Observation

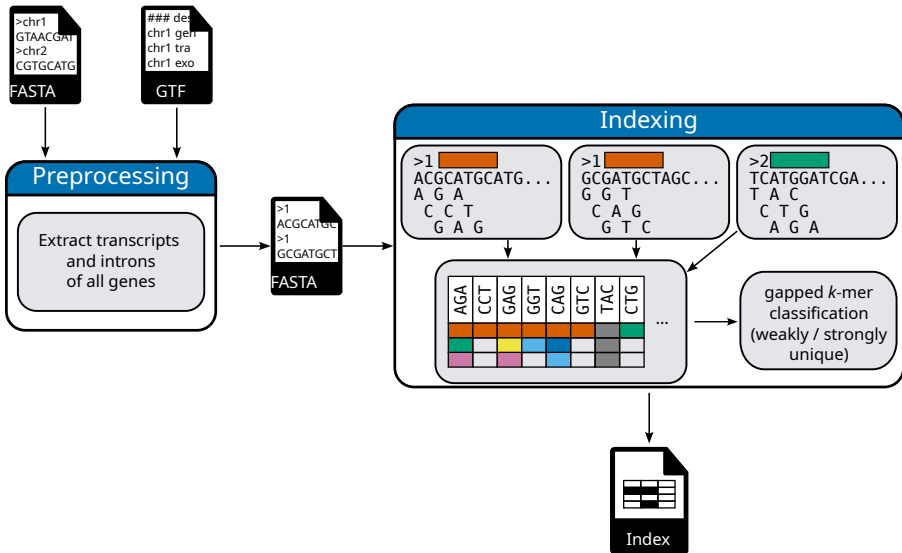
- Already high coverage using one color
- Most genes are covered using up to 3 colors

Idea

- Simplify colored de Bruijn graph by storing a limited number of colors per k -mer
- Depending on index: fewer cache misses per query



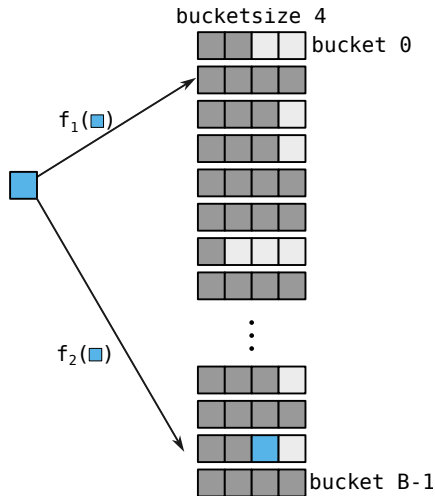
Arcane index



Arcane index

(2, 4)-Cuckoo Hashing

- We use $h = 2$ bucket hash functions $f_i : \mathcal{U} \rightarrow [B] := \{0, 1, \dots, B - 1\}$
- Each bucket has 4 slots
- To query a key, check if it is stored in one of the $h \cdot l$ slots
- To insert a key, check if one of $h \cdot l$ slots is empty, if not, perform a random walk
- Maximum achievable load for (2, 4)-Cuckoo hashing is ≈ 0.98

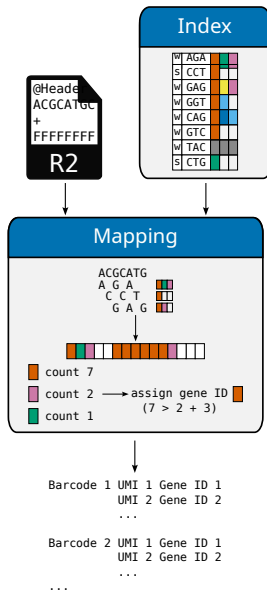


Walzer. Load Thresholds for Cuckoo Hashing with Overlapping Blocks. ACM Trans. Algorithms, 19(3):24:1–24:22, 2023. doi: 10.1145/3589558.

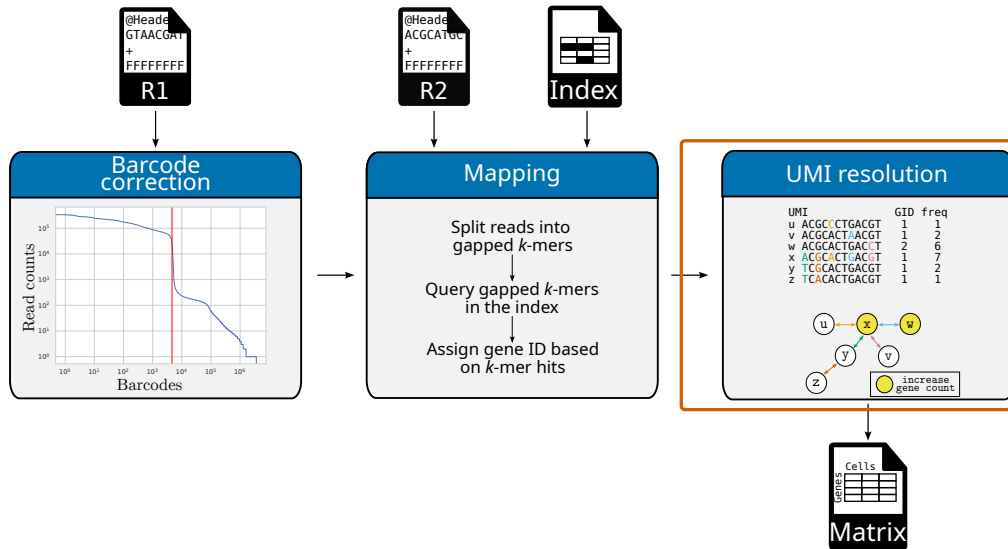
J. Zentgraf and S. Rahmann. Fast Gapped k -mer Counting with Subdivided Multi-Way Bucketed Cuckoo Hash Tables. WABI 2022, volume 242, pages 12:1–12:20, 2022. doi: 10.4230/LIPIcs.WABI.2022.12

Gene mapping

- For each read, query every gapped k -mer
- Count for each gapped k -mer its colors
- Gene weighting for more robust mapping:
 - Strongly unique k -mer: 5
 - Weakly unique k -mer: 3
 - Non-unique k -mer: 1
- Assign gene with highest count if its count is at least 3 larger than the next highest count



Overview of arcane

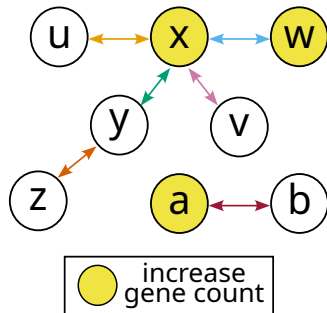


UMI resolution / de-duplication

Solution

For all UMIs with the same barcode

- Build a graph for all UMIs that are connected to their Hamming distance 1 neighbors
- Compute connected components in the graph
- For a connected component, count only UMI-gene combinations with a count $\geq$ the expected count



UMI		GID	freq
u	ACGC C CTGACGT	1	1
v	ACGCACT A ACGT	1	2
w	ACGCACTGAC C T	2	6
x	A CG C ACT G AC G T	1	7
y	T CGCACTGACGT	1	2
z	T CACTGACGT	1	1
a	ACTG A AACTGAG	7	5
b	ACTG T AACTGAG	7	2

Results

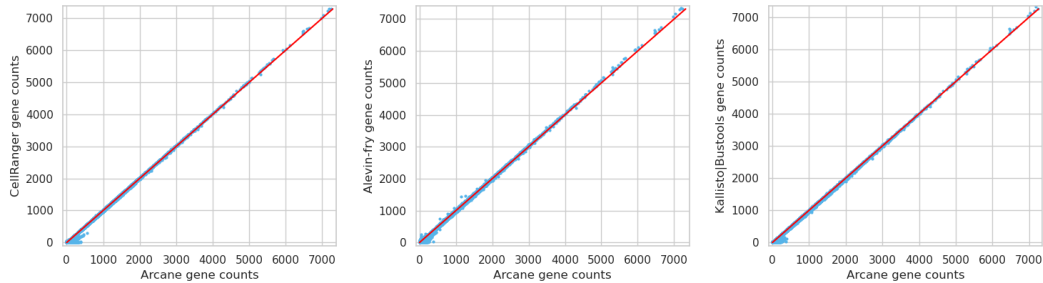
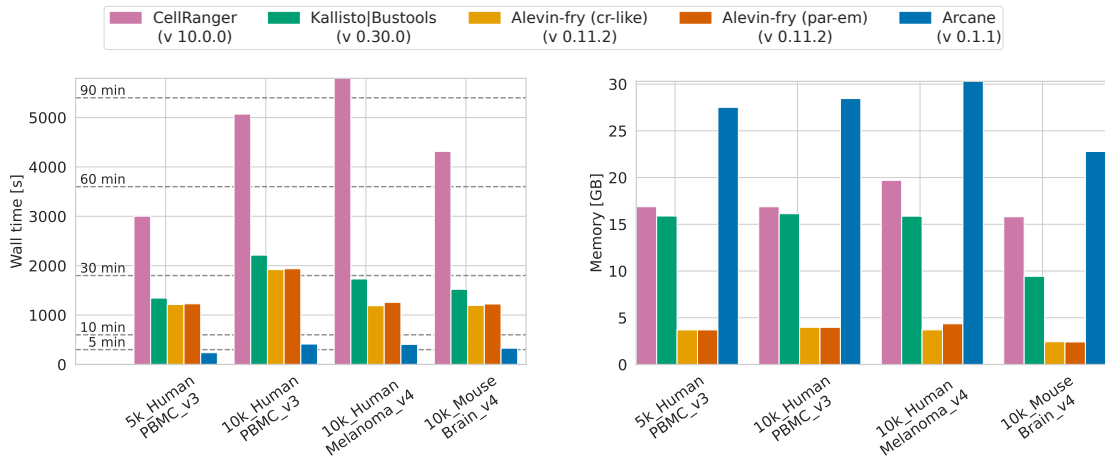


Figure: Comparison on the mouse brain dataset (10 000 cells, v4 chemistry).

Arcane index uses (31, 43) mask: ####_#_##_###_#_###_###_###_#_###_##_#_####

Results



All benchmarks were run on an Ubuntu (24.04 LTS) server with two AMD EPYC 9534 64-Core Processors, 1.5TB of 4800-MHz DDR5 memory and a KIOXIA CD8P SSD. Running time and maximum memory usage were measured using `/usr/bin/time -v` (using 16 threads).

Summary

Arcane

- Fast
- Memory footprint < 32 GB
- Supports shared memory for the index
- Yields similar results as Alevin-fry, Kallisto|Bustools and CellRanger

Ongoing work

- More pipelining
- Try out new index data structures to save memory

